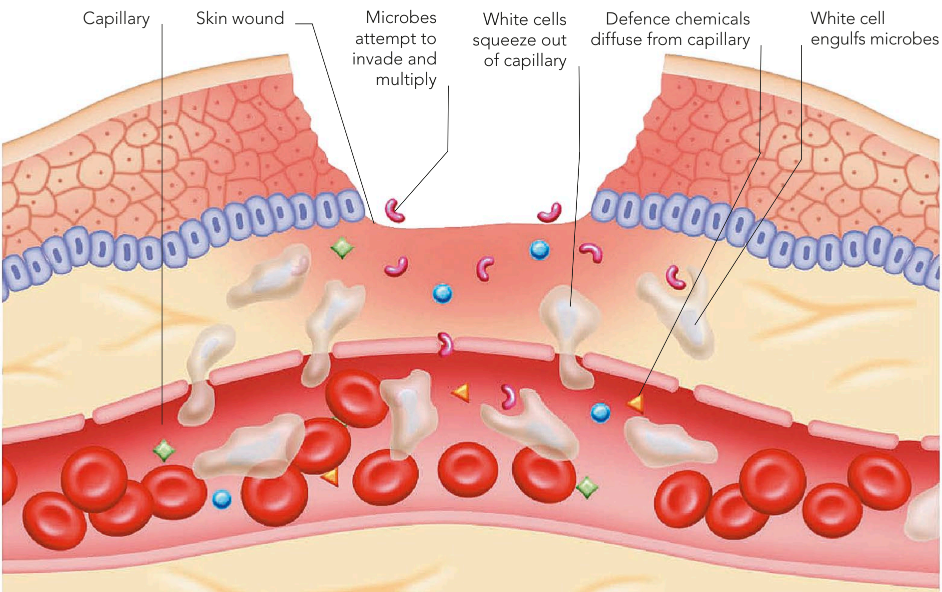
NON-SPECIFIC RESPONSE

Any damage, such as burns, extreme cold, corrosive chemicals, or invading organisms, elicits a non-specific response. The main response is inflammation (see p.198). The damaged tissue releases chemicals that attract white blood cells. Capillary walls become more permeable and porous to let these cells, along with defensive chemicals and fluids, enter and accumulate. The white cells surround, engulf, and destroy any pathogens, and the blood may clot to seal the breach.

INFLAMED TISSUE

The four common signs of inflammation are redness, swelling, increased warmth, and discomfort or pain. They occur after any form of harm in order to limit damage and initiate repair and healing.



LOCAL INFECTION

If harmful microbes enter body tissues, both the inflammatory and immune responses act swiftly to limit their spread. White blood cells, fluids, microbes, toxins, and debris accumulate as pus. An abscess forms if the pus gathers in a localized area, putting pressure on surrounding structures. This may cause discomfort and pain, especially if the surrounding tissues have no flexibility – for example, in a dental abscess.

DENTAL ABSCESS

Microbes enter through a region of decayed enamel and dentine, infect the pulp, and spread into the root, where pus collects. As pus presses on the pulp nerves, it causes the pain of toothache.